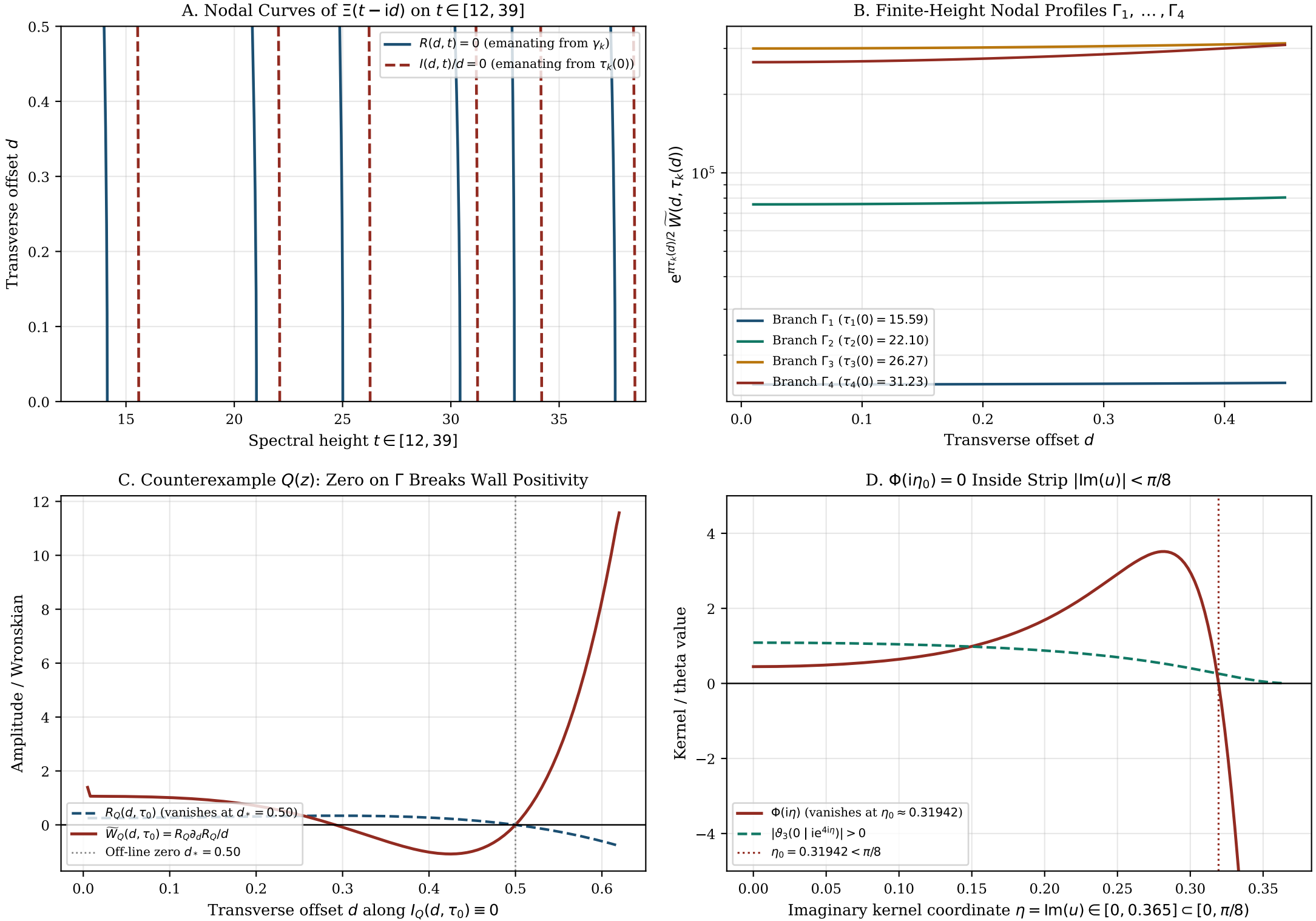
A. Nodal Curves of $\Xi(t - id)$ on $t \in [12, 39]$
Transverse offset d
Spectral height $t \in [12, 39]$
$R(d, t) = 0$ (emanating from γ_k)
$I(d, t)/d = 0$ (emanating from $\tau_k(0)$)
B. Finite-Height Nodal Profiles $\Gamma_1, \ldots, \Gamma_4$
$e^{\pi\tau_k(d)/2}\widetilde{W}(d, \tau_k(d))$
10^5
Transverse offset d
Branch Γ_1 ($\tau_1(0) = 15.59$)
Branch Γ_2 ($\tau_2(0) = 22.10$)
Branch Γ_3 ($\tau_3(0) = 26.27$)
Branch Γ_4 ($\tau_4(0) = 31.23$)
C. Counterexample $Q(z)$: Zero on Γ Breaks Wall Positivity
Amplitude / Wronskian
Transverse offset d along $I_Q(d, \tau_0) \equiv 0$
$R_Q(d, \tau_0)$ (vanishes at $d_* = 0.50$)
$\widetilde{W}_Q(d, \tau_0) = R_Q \partial_d R_Q / d$
Off-line zero $d_* = 0.50$
D. $\Phi(i\eta_0) = 0$ Inside Strip $|\mathrm{Im}(u)| < \pi/8$
Kernel / theta value
Imaginary kernel coordinate $\eta = \mathrm{Im}(u) \in [0, 0.365] \subset [0, \pi/8)$
$\Phi(i\eta)$ (vanishes at $\eta_0 \approx 0.31942$)
$|\vartheta_3(0 \mid ie^{4i\eta})| > 0$
$\eta_0 = 0.31942 < \pi/8$